

CAO MANH TAN

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EDUCATION

Ha Noi University of Science & Technology (HUST)

Bachelor of Computer Engineering

Ha Noi, Viet Nam

Sep 2022 – Present

Relevant Coursework: System Analysis and Design, Object-Oriented Programming (OOP), Data Structures and Algorithms, Database Systems, Web Technologies and Cloud Services.

WORK EXPERIENCE

TECHTACK LIMITED COMPANY

Software Engineer Intern

Ha Noi, Viet Nam

Feb 2026 – May 2026

- **ENSOGO – Internal Human Resources Management Web & App:**

- Built responsive admin dashboard interfaces and complex multi-criteria data tables with server-side pagination, sorting, and debounce search using **ReactJS**, **TypeScript**, and **Ant Design**.
- Implemented RESTful CRUD APIs using **Node.js** and optimized MySQL relational queries with composite indexing, cutting response latency by **40%** for large employee datasets.
- Developed reusable modal forms with strict client-side validation for leave management and staff onboarding workflows.

- **TechTack Internal – Time-Tracking & Attendance App:**

- Designed interactive weekly timesheet and attendance tracking views with real-time total hours calculation across desktop and mobile devices.
- Standardized **Axios** Request/Response Interceptors for centralized error handling and automated JWT token management; resolved 20+ peer code reviews following **Git Flow**.

Technologies & Tools: ReactJS, React Native, Expo, Node.js, NestJS, TypeScript, MongoDB, TypeORM, Ant Design, TanStack Query, Axios, Swagger, Git, Jira.

FEATURED PROJECTS

Hosting - Travel Stay & Hospitality Reservation Platform

Jun 2026 – Present

- **Domain-Driven Architecture:** Structured scalable system following Feature-Based frontend and DDD **NestJS 11** backend, guaranteeing clean separation between Controllers, Domain Services, and Repositories.
- **Concurrency Control:** Implemented Atomic Database Transactions using **typeorm-transactional**, completely eliminating race conditions and preventing double-booking during concurrent checkouts.
- **VietQR & Real-time SSE:** Integrated **payOS** dynamic QR generation with asynchronous Webhook listener (SHA-256 signature verification), streaming instant payment confirmation to clients via **Server-Sent Events (SSE)** and eliminating polling overhead.
- **Client Optimization & State:** Leveraged **TanStack Query v5** for automated server-state caching (slashing redundant requests by 60%), global **Zustand** for booking draft preservation, and **Leaflet Maps** with geocoding for homestay discovery.
- **Security & Auth:** Built Dual JWT authentication (Access/Refresh token) backed by a MySQL **Token Blacklist** table for immediate logout revocation, 3-tier RBAC, and client-side `<ProtectedRoute />` guards.

Technologies: ReactJS, NestJS 11, TypeScript, MySQL 8.0, TypeORM, TanStack Query v5, Zustand, payOS, SSE.

GoRide – Online Bus Booking & Fleet Management Platform

Mar 2026 – May 2026

- **Interactive Ticketing UX:** Developed dynamic 2D coach seat layout matrices (single/double-deck coaches) with real-time seat hold locking, visual availability legends, and dynamic price calculation.
- **Real-Time Customer Chat:** Engineered bidirectional customer support chat between passengers and operators using **Socket.IO** (NestJS WebSocket Gateway & React Context) with instant notifications.
- **High-Performance Data Layer:** Adhered to strict Scalar Foreign Keys conventions and custom TypeORM QueryBuilders for high-performance joins without eager-loading overhead, managing vehicle scheduling and financial settlements.
- **Authentication & Recovery:** Built 6-digit email OTP password recovery via **Resend API** with in-memory TTL caching, PayOS VietQR payment integration, and complete OpenAPI/Swagger 3.0 documentation.

Technologies: React 19, NestJS 11, TypeScript, MySQL 8.0, TypeORM, Socket.IO, Ant Design 6, Resend.

SKILLS & CERTIFICATIONS

Frontend: ReactJS (18/19), TypeScript, JavaScript (ES6+), HTML5, CSS3, SCSS, Ant Design (antd), Leaflet Maps, React Router v7.

State Management & Data Fetching: Zustand, TanStack React Query v5, Axios.

Backend: Node.js, NestJS 11, Express, TypeORM, Domain-Driven Design (DDD) Architecture, RESTful APIs, Server-Sent Events (SSE / EventSource), WebSocket / Socket.IO.

Database & ORM: MySQL 8.0, MongoDB, Custom QueryBuilder, Atomic Transactions.

Security & Integrations: Dual JWT (Access/Refresh), Token Blacklisting, RBAC, SHA-256 Webhook, payOS (VietQR), Cloudinary, Resend, Nodemailer.

Tools: Git, GitHub, Vite, Postman, Swagger / OpenAPI 3.0, Cronjob, Jira.

Languages: Vietnamese (Native), English (Reading Technical Documentation, Basic Communication).

Certifications: TOEIC 755/990 (Listening & Reading).